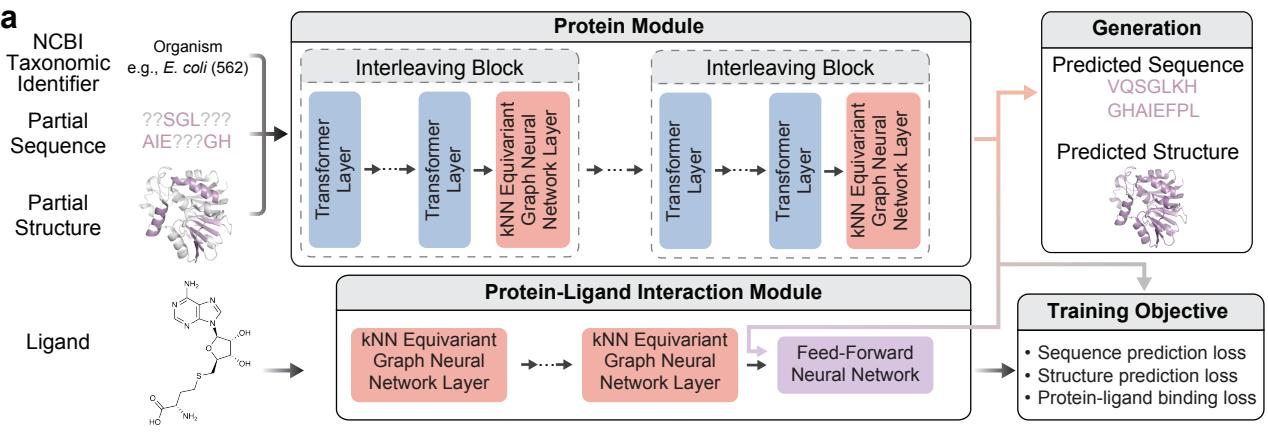
a
NCBI Taxonomic Identifier
Organism
e.g., E. coli (562)
Partial Sequence
??SGL???
AIE???GH
Partial Structure
Ligand
Protein Module
Interleaving Block
Transformer Layer
Transformer Layer
kNN Equivariant Graph Neural Network Layer
Interleaving Block
Transformer Layer
Transformer Layer
kNN Equivariant Graph Neural Network Layer
Protein-Ligand Interaction Module
kNN Equivariant Graph Neural Network Layer
kNN Equivariant Graph Neural Network Layer
Feed-Forward Neural Network
Generation
Predicted Sequence
VQSGLKH
GHAIEFPL
Predicted Structure
Training Objective
• Sequence prediction loss
• Structure prediction loss
• Protein-ligand binding loss